

POWERFACTORY

PowerFactory 2024

Technical Reference

PV MPP Tracking Controller

ElmPvmpp

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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Contents

1 General Description **1**

1.1 Maximum Power Point of Single PV Panel 1

1.2 Maximum Power Point of Partially Shaded PV String 1

1.3 Solar Power System Topology with MPP Tracker 3

1.4 Common MPP Tracking Algorithms 3

2 MPPT Controller Setup **4**

2.1 Input of PV Array Topology 4

2.2 Basic Data 5

3 Load Flow Analysis **5**

3.1 Analytic MPP Pre-Calculation 5

3.2 MPP Tracking in Outer Loop 5

4 RMS and EMT Simulation **6**

4.1 Input parameters 6

4.2 Perturb and Observe Algorithm 7

4.3 Incremental Conductance Algorithm 7

5 References **9**

List of Figures **10**

List of Tables **11**

1 General Description

This documents describes the maximum power point tracker (MPPT) model *ElmPvmpp*. It is a controller for converters DC-DC or PWM-converters (*ElmDcdc*, *ElmVsc*) and can also be applied to the DC voltage source with two terminals (*ElmDcubi*). It is required for usage of the PV panel model (*ElmPvpanel*).

1.1 Maximum Power Point of Single PV Panel

Solar cells show a non-linear current-voltage characteristic specified by $I(U)$ curves. As a result, there is a point of maximum power (MPP) characterised by a specific voltage U_{MPP} and a corresponding current I_{MPP}. The MPP depends on the actual irradiation on the PV panel surface and the cell temperature. An example for irradiation dependent $I(U)$ and corresponding $P(U)$ curves is given in Figure 1.1, where the MPPs are marked with circles.

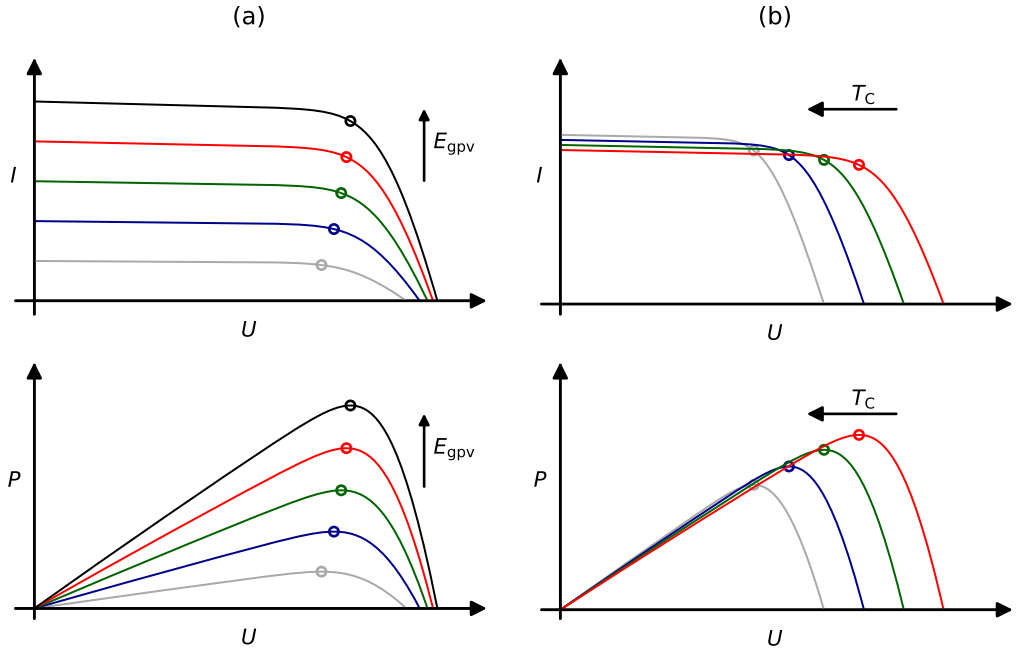


Figure 1.1: Exemplary $I(U)$ and $P(U) = I(U) \cdot U$ curves for (a) different irradiations $E_{\mathrm{g,pv}}$ and (b) different cell temperatures T_{C}

1.2 Maximum Power Point of Partially Shaded PV String

If multiple PV panels of same type, irradiation and cell temperature are connected in series to a string, all panels will have the same current while the voltages just add up. The $I(U)$ and $P(U)$ characteristics will have the same shape as a single panel but voltage axis will be multiplied by the number of panels.

But in case the modules differ, e.g. due to partial shading of a string, the current will be limited by the panel with the smallest photovoltaic current.

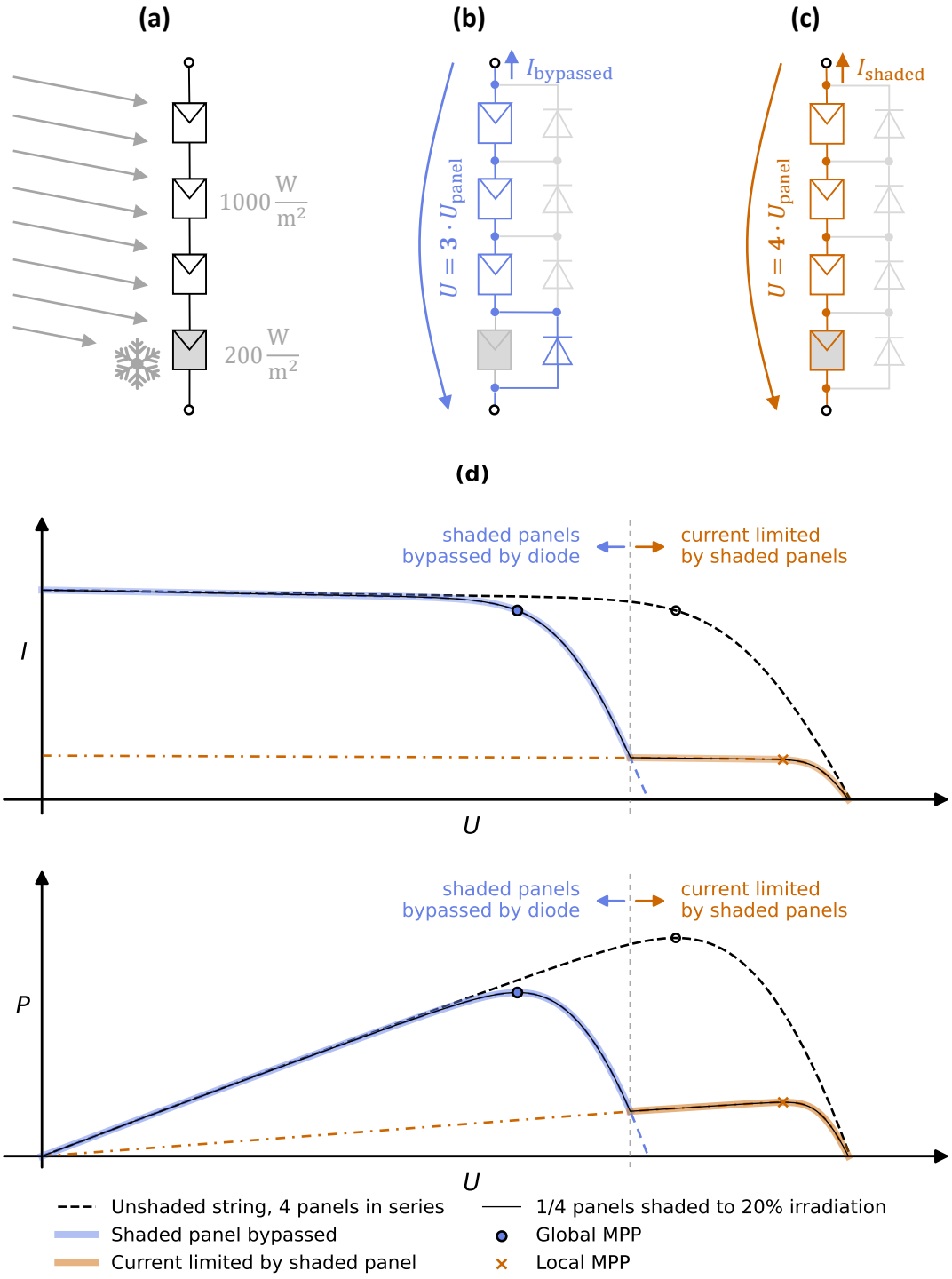


Figure 1.2: Example of partially shaded solar generator string,

(a) Different irradiations due to shading, e.g. caused by snow accumulated on bottom panel
(b) Current path through bypass diode with reduced voltage and higher current
(c) Current path through shaded panel with higher voltage and reduced current
(d) $I(U)$ and $P(U)$-curves

To protect the solar cells, most panels have integrated bypass diodes. These will come into effect, if the string voltage is lowered sufficiently, basically removing the affected cells/panels from the string since they are bypassed.

The resulting $I(U)$ and $P(U)$ curves will look quite different, as can be seen in the Figure 1.2. These apply for a string of four PV panels in series with one panel being shaded to about 20% of the other modules irradiation.

For high voltages all PV panels are active in the string and the current is limited by the shaded panels. For lower voltages the current suddenly increases to the values of the not shaded modules. Therefore two local maxima in the $P(U)$ characteristic occur. In this case, the maximum at the lower voltage is higher due to the significant current increase.

Note: Since the bypass diodes are essential for the described behaviour, by default a bypass diode model is included in the PV panel model. It can be removed by de-selecting *Include bypass diode into circuit model* on the Load Flow page of the panel.

1.3 Solar Power System Topology with MPP Tracker

Figure 1.3 shows a typical PV topology including a PV panel, a DC/DC boost converter and an inverter connected to the grid. The PV panel can also represent multiple serial PV panels connected to a string or an array of multiple parallel strings. In this topology the DC/DC-converter is responsible to provide a DC voltage according to the MPP of the connected PV array.

This functionality is referred to as MPP tracking and is done by the MPPT controller. The controller uses the measured voltage and current to adapt the duty cycle of the converter to maximize the power output.

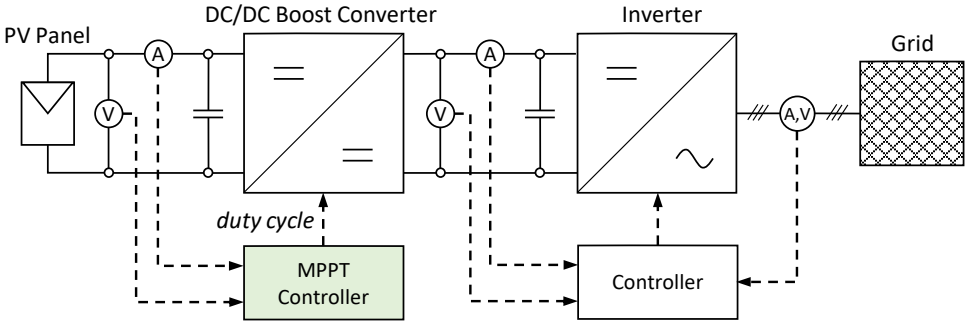


Figure 1.3: Schematic of PV generator, DC-DC boost converter with MPP controller and inverter connected to the power grid

1.4 Common MPP Tracking Algorithms

MPP tracking algorithms work by incrementally changing the DC voltage over time and analysing the power output [1]. Two commonly used algorithms are

- Perturb and Observe (P&O)
- Incremental Conductance (IC)

While the P&O algorithm always alternates around the MPP, changing the voltage in every step, the IC algorithm stays at a constant voltage operating point once the MPP is found. Only if the output current changes due to external factors, such as varying temperature or irradiation, the new MPP is searched by voltage variation.

2 MPPT Controller Setup

2.1 Input of PV Array Topology

As described in section 1.4, MPP tracking algorithms work by incrementally changing the DC voltage. Therefore they adopt multiple points on the $I(U)$ curve over time.

This is not possible in stationary calculations such as the Load Flow Analysis or the initialisation of time domain simulations. Instead different points on the I(U) curve can be approached in outer loop iterations. But to completely span the entire range of possible voltages would take a high number of iterations, which could drastically increase the calculation time. To avoid this issue, the $I(U)$ and $P(U)$ curves are pre-calculated analytically to approximate the MPP voltage. This approximation is then used as a starting point for outer loop iterations, to find the actual MPP more precisely.

To pre-calculate $I(U)$ and $P(U)$ analytically, the MPP tracker model needs detailed information about the PV panel topology, i. e. which panels are connected in which way on the DC side of the controlled converter.

The following information is provided by the user in the *Contents* of the MPPT model:

- Each PV string (at least one) is defined as Vector of Objects (*IntVecobj*)
- Each PV panel (at least one per string) is added to a PV string by adding a row in the corresponding Vector of Objects and assigning the PV panel to the first cell

Figure 2.1 shows how to insert the PV topology information in the *Contents* of the MPPT controller for three parallel strings, each containing n Panels.

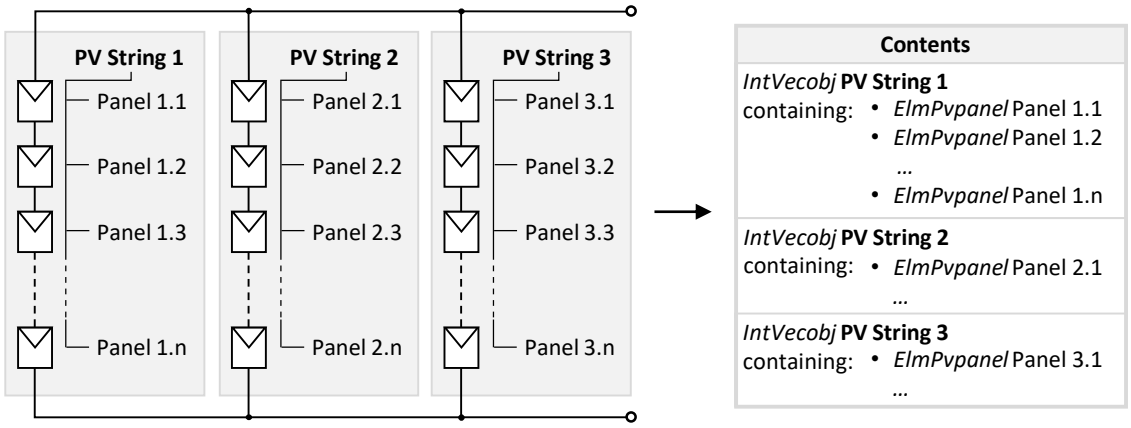


Figure 2.1: PV panel topology as input in *Contents* of MPPT controller

2.2 Basic Data

The parameters on the Basic Data page of the MPPT controller are shown in Table 2.1.

Table 2.1: Input parameters in Basic Data Page

Parameter	Unit	Default Value	Description	Symbol
pConv			Controlled converter / DC element	
usetp_mx	p.u.	1.1	Upper voltage limit	
usetp_mn	p.u.	0	Lower voltage limit	

The *Controlled converter / DC element* can either be a DC/DC Converter (*ElmDcdc*), a PWM Converter (*ElmVsc*) or a DC Voltage Source (*ElmDcubi*). The voltage limits are per unit values of the nominal DC voltage of the controlled element and are used for the MPP pre-calculation.

3 Load Flow Analysis

3.1 Analytic MPP Pre-Calculation

The MPP voltage is estimated analytically using the known PV panel topology and the panel parameters. This returns an approximation of the entire $I(U)$ and $P(U)$ curves and several relevant parameters, such as

- the estimated global MPP voltage $U_{\mathrm{MPP,PC}}$,
- the estimated global MPP power $P_{\mathrm{MPP,PC}}$ and
- the estimated minimum distance between local maxima $\Delta U_{\mathrm{locMPP,PC}}$.

The estimated voltage is only an approximation of the actual MPP voltage. Several assumptions and simplifications are made, such as:

- Voltage drops over lines, resistors and other elements within the PV topology are neglected.
- Elements parallel to the PV strings are neglected.
- Circulating currents between strings due to voltage mismatch are neglected.

To obtain a more precise value for the MPP voltage, an MPP tracking algorithm is applied in the Outer Loop.

3.2 MPP Tracking in Outer Loop

After the initial Load Flow calculation the actual $P(U_{\mathrm{MPP,PC}})$ is compared to $P_{\mathrm{MPP,PC}}$. Differences between pre-calculated and actual value smaller than $5\,\%$ are interpreted as good accordance. In this case a second voltage point only 0.01 p.u. left of the initial value is adopted in the next iteration. The relation $\Delta P/\Delta U$ is used as fast exit criterion to check if the MPP is actually found.

If the check is negative or the initial difference is larger than $5\,\%$, the next task is to find the voltage interval containing the actual MPP. Therefore points around $U_{\mathrm{MPP,PC}}$ are adopted with outer loop iterations. The voltage step size for this task is chosen based on the distance $\Delta U_{\mathrm{locMPP,PC}}$ of local maxima on the pre-calculated $P(U)$ curve. The search interval is found when the highest known point has two neighbours with lower corresponding P values.

The next task is to find the actual maximum within the search interval. Initially an interval bisection method is used, alternately intersecting the intervals right and left of the currently known maximum. This method is relatively robust but takes a higher number of iterations to precisely find the maximum. Therefore the search method is changed, if the MPP is narrowed down to a range smaller $\Delta U_{\mathrm{locMPP,PC}}$, in which several consecutive points follow the condition $\frac{\mathrm{d}^2P}{\mathrm{d}U^2} < 0$. Henceforth a cubic spline interpolation is performed on the known points to get the next estimation for U_{MPP}. This method takes very few iterations but is only reliable close to the actual MPP.

Different exit criteria are established to facilitate a fast and reliable determination of whether the MPP has been found:

- Very fast exit criterion: Requires $\geq$ 2 known points, usually applied if pre-calculated MPP is close to actual MPP, based on $\Delta P/\Delta U$ of two highest known points
- Fast exit criterion: Requires $\geq$ 3 known points, based on voltage distance between neighbours of highest known point
- Robust exit criterion: Requires $\geq$ 4 known points, checks if there are significant changes in the last 4 iterations

4 RMS and EMT Simulation

The existing P&O and IC MPP tracking algorithms are be implemented for RMS and EMT simulations. Therefore the voltage and current at the terminal of the controlled element are measured and filtered using first order lags (PT1-filter, time constant τ_{F}) and then used to analyse the effect of incremental voltage setpoint changes.

4.1 Input parameters

Both algorithms need the input parameters given in table 4.1. Some additional parameters for the IC algorithm are required acc. to table 4.2. The per unit values are nominalised based on the nominal DC voltage $U_{\mathrm{nom,DC}}$ of the controlled element and the pre-calculated MPP power $P_{\mathrm{MPP,PC}}$.

Table 4.1: Input parameters for RMS and EMT simulation for both algorithms

Parameter	Unit	Default	Description	Range	Symbol
algorithm		0 (P&O)	Algorithm	0, 1	
tclock	ms	1 ms	Update time step	> 0	t_{clock}
deltaU	p.u.	0.01	Voltage step	> 0	Δu
Tf	ms	0.3	Input filter time constant	> 0	τ_{F}

Table 4.2: Additional input parameters for RMS and EMT simulation for IC algorithm

Parameter	Unit	Default	Description	Range	Symbol
u_thresh	p.u. (of Δu)	0.1	Voltage change	> 0	u_{thresh}
i_thresh	p.u.	0.05	Current change	> 0	i_{thresh}

4.2 Perturb and Observe Algorithm

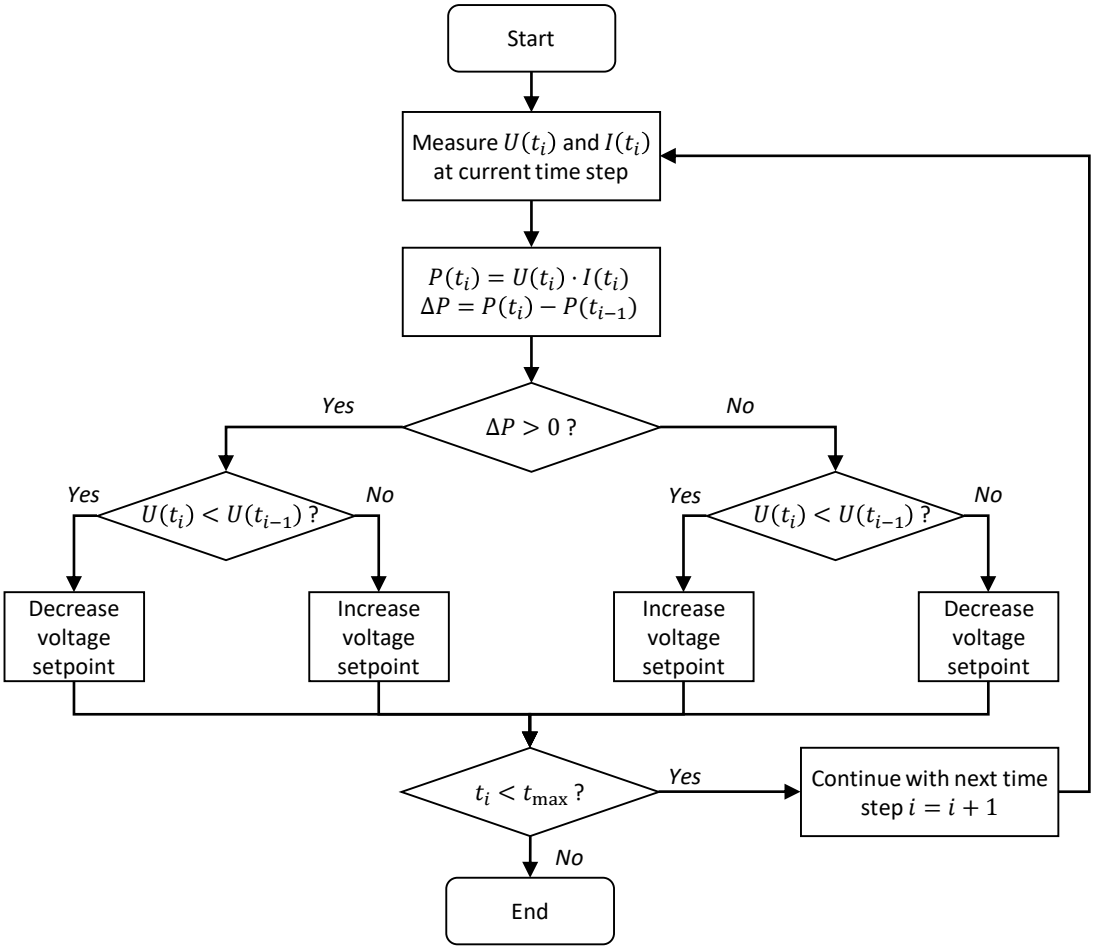


Figure 4.1: Flowchart of Perturb and Observe (P&O) algorithm

4.3 Incremental Conductance Algorithm

The flow chart of the IC algorithm is shown in Figure 4.2. The threshold values ε_{U} and ε_{I} are calculated from the input parameters as follows:

$$\varepsilon_{\mathrm{U}} = U_{\mathrm{nom,DC}} \cdot \Delta u \cdot u_{\mathrm{thresh}} \tag{1}$$

$$\varepsilon_{\mathrm{I}} = \frac{P_{\mathrm{MPP,PC}}}{U_{\mathrm{nom,DC}}} \cdot i_{\mathrm{thresh}} \tag{2}$$

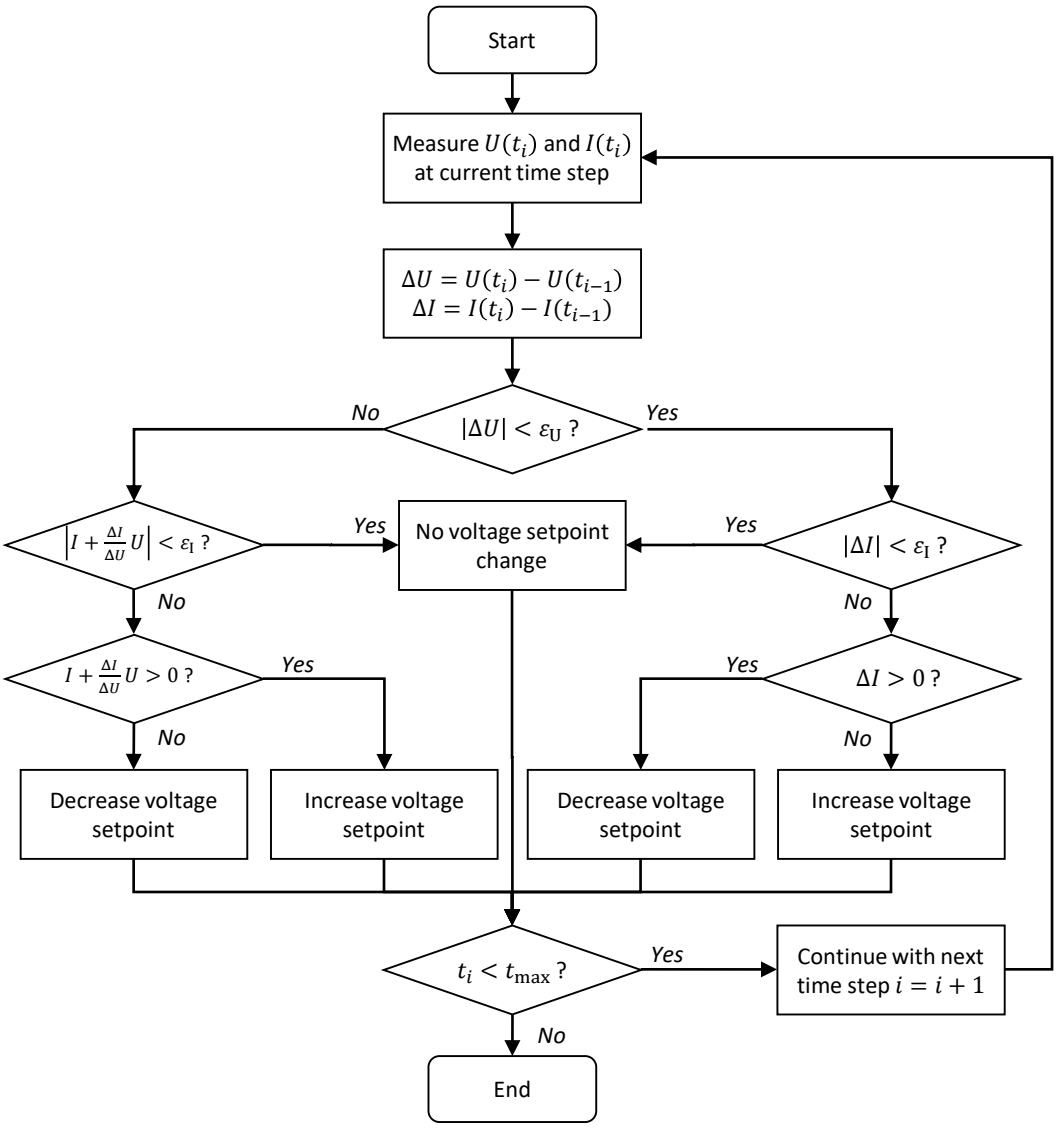


Figure 4.2: Flowchart of Incremental Conductance (IC) algorithm

5 References

[1] Z. M. S. Elbarbary and M. A. Alranini. Review of Maximum Power Point Tracking Algorithms of PV System. *Frontiers in Engineering and Built Environment*, 1:68–80, 2021.

List of Figures

1.1 Exemplary $I(U)$ and $P(U) = I(U) \cdot U$ curves for (a) different irradiations $E_{\mathrm{g,pv}}$ and (b) different cell temperatures T_{C} 1

1.2 entry for lot 2

1.3 Schematic of PV generator, DC-DC boost converter with MPP controller and inverter connected to the power grid 3

2.1 PV panel topology as input in *Contents* of MPPT controller 4

4.1 Flowchart of Perturb and Observe (P&O) algorithm 7

4.2 Flowchart of Incremental Conductance (IC) algorithm 8

List of Tables

2.1 Input parameters in Basic Data Page 5

4.1 Input parameters for RMS and EMT simulation for both algorithms 6

4.2 Additional input parameters for RMS and EMT simulation for IC algorithm 7